TECHNICAL & STRATEGY REPORT

EcoCollecte Bangui

Civic Cloud Platform for Urban Waste & Sanitation Tracking

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1. INTRODUCTION & PROBLEM STATEMENT

EcoCollecte Bangui is a web application designed to empower citizens to report illegal dumpsites and sanitation emergencies across their neighborhoods. It provides the City Hall of Bangui with a modern, cloud-native tool to track every issue from initial report to complete resolution, provided as a **100% free civic contribution** to the Central African population.

The underlying issue is primarily **informational**. Residents witnessing overflowing waste or clogged drainage channels had no simple public medium to alert city services. Consequently, municipal authorities lacked a consolidated, real-time map of waste accumulation, leading to blind planning for collection rounds.

3-Click Reports
FRENCH, SANGO, ENGLISH

Agent Workspace
SECURE STATUS TRACKING

Transparency
LIVE PUBLIC COUNTERS

Free Access
OFFERED TO THE NATION

2. CONTEXT: WASTE & SANITATION IN BANGUI

Bangui produces far more solid waste than city services can collect. According to the **UN-Habitat** urban profile, the capital houses approximately 1.5 million inhabitants across 67 km² (divided into 8 arrondissements). Daily waste production is estimated at **930 m³**, compared to only **320 m³ actually collected**.

Three Parallel Waste Collection Chains

- **Informal collectors:** Operate with pushcarts, frequently depositing waste in rainwater channels or vacant lots.
- **City Hall (Mairie):** Focuses limited equipment on administrative areas and commercial centers.
- **HYSACA (Private Operator):** Holds direct household contracts in central and selected working-class districts.

The city operates 41 transfer stations, 12 skip-handler trucks, and employs roughly 250 temporary street sweepers. The Kolongo landfill is reaching end-of-life capacity, while uncollected waste blocking drainage canals severely increases seasonal flooding risks.

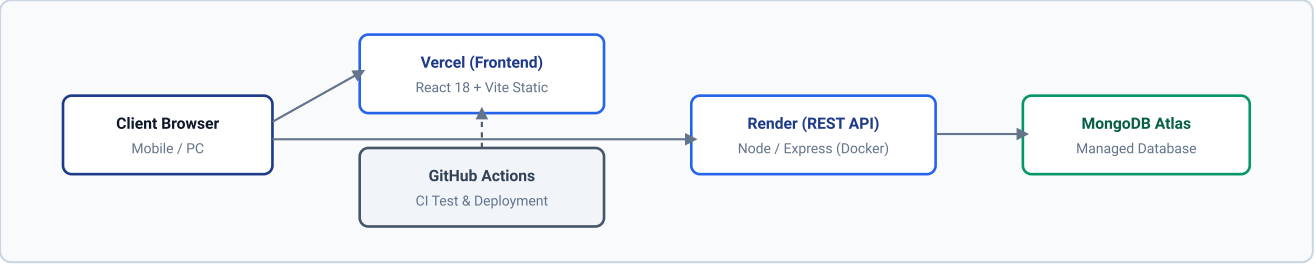
3. SOLUTION ARCHITECTURE & FEATURES

EcoCollecte is structured as a Single-Page Web Application (SPA) connected to a REST API. Citizens do not need an account to report issues, removing barriers to public adoption. Protected agent access allows municipal staff to update issue statuses securely.

MODULE	CORE FUNCTIONALITY	TARGET AUDIENCE
Home Page	Platform presentation, instant report button, public awareness banner, live counters.	General Public
Reports Portal	Structured submission form (category, description, district) and filterable public feed.	Residents
Agent Portal	Protected dashboard allowing staff to update status: *Reported* → *In Progress* → *Resolved*.	City Hall Agents
News & Media	Bangui photo gallery, civic educational articles, and official sanitation notices.	All Users

4. CLOUD ARCHITECTURE & CI/CD PIPELINE

The system relies on a three-tier architecture hosted on reliable cloud infrastructures, backed by an automated continuous integration pipeline.



Core Design Principles

1. **Stateless API:** Render container retains no local state; all records persist in MongoDB Atlas.
2. **Complete Decoupling:** Frontend manages interface rendering; backend handles validation and logic.
3. **Secret Management:** Sensitive database credentials and admin keys are passed via secure environment variables.

5. TECHNOLOGY STACK

LAYER	TECHNOLOGY	TECHNICAL JUSTIFICATION
Frontend	React 18 + Vite 5	Optimized static bundle, fast rendering, highly responsive form handling.
Localization	Native i18n (FR, SG, EN)	Lightweight React translation context without heavy third-party overhead.
Backend API	Node.js 20 + Express 4	Unified JavaScript ecosystem, lightweight and well-documented runtime.
Validation	Zod Schema	Ensures strict request body validation and filters out malformed payloads.
Database	MongoDB Atlas + Mongoose 8	Document store ideal for flexible report structures and public notices.
Container	Docker	Guarantees identical API runtime across local development and cloud production.
Hosting	Vercel & Render	Continuous cloud deployment directly triggered by Git repository pushes.
CI/CD & Testing	GitHub Actions, Vitest, Supertest	13 automated unit and integration tests run on every push (avg duration: 22s).

6. CORE API ENDPOINTS & CI PIPELINE

HTTP ROUTE	FUNCTION	ACCESS & SECURITY
`GET /health`	API health check invoked by cloud runtime	Public
`GET /api/reports`	Fetch reports list (filterable by status)	Public
`POST /api/reports`	Submit a new citizen report	Public (Zod Schema Validated)
`PATCH /api/reports/:id/status`	Update report status (e.g. to resolved)	Municipal Agents (`x-admin-key` header)

7. SECURITY MEASURES

- **Secret Isolation:** Zero hardcoded credentials in source control; template `.env.example` provided.
- **Constant-Time Verification:** Admin keys compared using `timingSafeEqual` to mitigate side-channel timing attacks.
- **Rate Limiting & Payload Caps:** Restricted via `express-rate-limit` (200 req / 15 min per IP) with a 10 KB body cap.
- **Security Headers:** Helmet integration and tight CORS configuration permitting calls only from official domain instances.

8. OPERATIONAL LIMITS & MITIGATION STRATEGIES

TECHNICAL LIMIT	OBSERVED IMPACT	REMEDIATION PLAN
API Cold Starts	Initial request delay after periods of inactivity.	Implement periodic ping requests to keep API warm.
Shared Agent Key	No individual audit log for status updates.	Deploy nominative agent accounts with activity logs.
Text-Only Reports	Agents cannot visually verify dump size.	Integrate Cloudinary for lightweight photo attachments.
Spam Prevention	Potential risk of fraudulent entries.	Add CAPTCHA verification and moderation queue.

9. PROJECT ROADMAP

Phase 1: Feature Enhancements

- Photo upload integration (Cloudinary storage).
- Automated citizen status notifications via email.
- Individual agent authentication with action logs.

Phase 2: Municipal Deployment

- Interactive map across the 8 arrondissements.
- Pilot deployment in two trial arrondissements.
- Data moderation and privacy policy adoption.

10. CONCLUSION

EcoCollecte Bangui achieves all its core objectives. By providing a modern, multilingual, and accessible platform, it establishes a transparent digital bridge between the residents of Bangui and municipal services. Delivered as a **free public utility**, it gives local authorities the crucial data layer required to optimize urban waste management and protect public health.

Sources & Institutional References

• UN-Habitat: Bangui Urban Profile (Waste volumes, municipal equipment, collection chains).
• African Development Bank / AWF: PERISA-GB Mission Appraisal Report (March 2025).
• Cloud & CI Dashboards: Operational metrics derived from Vercel, Render, and GitHub Actions.